

Functional Requirements Document

ChurnGuard — Churn-Risk Detection & Retention Workflow

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1. Purpose and audience

This document states what ChurnGuard must do, in system terms, to satisfy the business requirements in BRD-CG-001. It is written for the delivery team, QA and UAT participants. Design-level detail — algorithms, field types, screen behaviour — is in FSD-CG-001. Non-functional requirements are in SRS-CG-001.

Each requirement is atomic, testable and traced to its parent BR.

Convention: *shall* = mandatory. Every requirement carries a verification method: **T** (test), **D** (demonstration), **I** (inspection), **A** (analysis).

2. Actors

Actor	Description
CSM	Customer Success Manager owning one or more high-value accounts
CS Manager	Manages a CSM team; reviews escalations and team risk position
VP Customer Success	Consumes portfolio-level view; approves threshold changes
Business Administrator	CS Ops role authorised to change weights, thresholds and SLAs
Scoring Engine	Scheduled system process that ingests, scores and classifies
Source Systems	Usage data mart, marketplace connector service, support desk, billing

3. Functional requirements

3.1 Data ingestion and preparation — traces to BR-01, BR-02, BR-11

ID	Requirement	Priority	Verify	Parent
FR-001	The system shall retrieve, once per scoring cycle, product usage metrics per account: active days, orders processed, listings published, distinct modules used — for the trailing 30-day and trailing 90-day windows.	Must	T	BR-02
FR-002	The system shall retrieve marketplace connector health per account: connection state, last successful sync timestamp, sync error count and sync attempt count for the trailing 30 days, and any marketplace account-health warning flag.	Must	T	BR-02
FR-003	The system shall retrieve support metrics per account for the trailing 30 and 90 days: ticket count, count by priority, SLA breach count, reopened ticket count, and most recent CSAT score.	Must	T	BR-02
FR-004	The system shall retrieve commercial data per account: subscription plan and seat count with prior values, failed payment attempts in the trailing 60 days, oldest overdue invoice age, and next renewal date.	Must	T	BR-02
FR-005	The system shall retrieve engagement data per account: date of last logged meaningful activity, QBR attendance in the trailing two quarters, primary contact active status, and outbound-to-response ratio for the trailing 90 days.	Must	T	BR-02
FR-006	The system shall resolve every source record to a single CRM Account ID and shall log any record it cannot resolve to an exception report without halting the cycle.	Must	T	BR-11
FR-007	The system shall mark a signal as <i>Insufficient Data</i> where its source feed is older than 48 hours or where more than 20% of that signal's required inputs are null for the account.	Must	T	BR-11
FR-008	The system shall suppress composite scoring for an account where two or more of the five signals are marked <i>Insufficient Data</i> , retain the previous score, and set the account's data-quality flag to <i>Suppressed</i> .	Must	T	BR-11
FR-009	Where exactly one signal is marked <i>Insufficient Data</i> , the system shall redistribute that signal's weight proportionally across the remaining signals and record that the score was computed on a reduced signal set.	Must	T	BR-11

3.2 Signal scoring — traces to BR-01, BR-02

ID	Requirement	Priority	Verify	Parent
FR-010	The system shall calculate a sub-score of 0–100 for each of the five signals, where 0 represents no risk and 100 represents maximum risk, using the banded rules defined in FSD-CG-001 §3.	Must	T	BR-02
FR-011	The system shall calculate a composite risk score as the weighted sum of the five sub-scores, rounded to a whole number in the range 0–100.	Must	T	BR-01
FR-012	The system shall record, for every scored account, the sub-score, the raw input values and the comparison period used for each signal.	Must	T	BR-06
FR-013	The system shall identify the dominant risk driver as the signal contributing the highest weighted points to the composite score, and shall record the two next-highest contributors.	Must	T	BR-05
FR-014	The system shall complete a full scoring cycle for all in-scope accounts once per day within the agreed batch window.	Must	T	BR-01
FR-015	The system shall permit an authorised user to trigger an on-demand rescore of a single account, and shall complete it within the interactive response time stated in SRS-CG-001.	Should	D	BR-01

3.3 Risk tiering and change detection — traces to BR-03

ID	Requirement	Priority	Verify	Parent
FR-016	The system shall assign a risk tier from the composite score using configured boundaries: Stable, Watch, Critical.	Must	T	BR-03
FR-017	The system shall compare the newly assigned tier with the tier from the previous cycle and classify the movement as No Change, Escalation or De-escalation.	Must	T	BR-03
FR-018	The system shall apply a configurable hysteresis buffer of ± 3 points at each tier boundary, so that an account does not change tier on a movement smaller than the buffer.	Should	T	BR-03
FR-019	The system shall write a dated score-history record for every account on every cycle, containing composite score, each sub-score, tier, movement type and data-quality flag.	Must	T	BR-09
FR-020	The system shall retain score history for 13 months and shall archive rather than delete records beyond that period.	Must	I	BR-09

3.4 Alerting, task routing and SLA — traces to BR-04, BR-05

ID	Requirement	Priority	Verify	Parent
FR-021	On an Escalation into Watch or Critical, the system shall create a Retention Task assigned to the account's owning CSM.	Must	T	BR-04
FR-022	The Retention Task shall carry: account name, composite score, previous and new tier, dominant risk driver, the top three contributing signals with their values, and a link to the account risk detail view.	Must	T	BR-04, BR-06
FR-023	The system shall set the task due date from the configured SLA for the new tier — Critical 1 business day, Watch 3 business days — calculated against the CS working calendar.	Must	T	BR-04
FR-024	On an Escalation from Stable directly to Critical in a single cycle, the system shall additionally notify the CSM's CS Manager.	Must	T	BR-04
FR-025	Where the account has no active owning CSM, the system shall assign the task to the CS Manager queue for the account's team and flag the ownership gap.	Must	T	BR-04
FR-026	The system shall not create a second Retention Task for an account with an open task for the same tier; it shall instead update the existing task with the new score and append a note.	Must	T	BR-04
FR-027	The system shall attach the retention playbook mapped to the dominant risk driver, taking the mapping from a business-maintained configuration table.	Must	T	BR-05
FR-028	The system shall record first-touch timestamp when the owning CSM logs the first activity against the account after task creation, and shall calculate elapsed business time from task creation to first touch.	Must	T	BR-04, BO-4
FR-029	The system shall mark a task as SLA Breached where the due date passes without a recorded first touch, and shall notify the CS Manager.	Must	T	BR-04
FR-030	The system shall suppress alert generation for any account during a configured shadow-run period, while continuing to score and store history.	Must	D	R-1 mitigation

3.5 Risk detail view — traces to BR-06

ID	Requirement	Priority	Verify	Parent
FR-031	The system shall present, on the Account record, a risk panel showing composite score, tier, movement since last cycle, dominant driver and last scored timestamp.	Must	D	BR-06
FR-032	The risk panel shall show each of the five signals with its sub-score, weighted contribution, the underlying input values and the comparison period.	Must	D	BR-06
FR-033	The risk panel shall display a 90-day score trend line for the account.	Should	D	BR-09
FR-034	Where scoring was suppressed or computed on a reduced signal set, the panel shall display the data-quality status and which signals were unavailable.	Must	D	BR-11
FR-035	The risk panel shall provide a single action to open a Retention Case pre-populated from the current score and dominant driver.	Must	D	BR-07

3.6 Retention case management — traces to BR-07

ID	Requirement	Priority	Verify	Parent
FR-036	The system shall provide a Retention Case record linked to one Account, holding: opening score and tier, dominant driver, playbook applied, owner, status, actions taken, closure outcome and closure reason.	Must	T	BR-07
FR-037	The system shall support case statuses: Open, In Progress, Awaiting Customer, Closed–Saved, Closed–Churned, Closed–False Positive.	Must	T	BR-07
FR-038	The system shall require a closure reason from a controlled list before a case may be set to any Closed status.	Must	T	BR-07

FR-039	The system shall prevent more than one open Retention Case per account at a time.	Must	T	BR-07
FR-040	The system shall automatically close a case as Closed-Saved where the account's tier returns to Stable and remains Stable for 30 consecutive days, unless the case owner has set a manual hold.	Should	T	BR-07
FR-041	The system shall link the Retention Case to the renewal outcome recorded on the account so that saved and churned results can be reported against opening tier.	Must	T	BR-14

3.7 Dashboard and reporting — traces to BR-08

ID	Requirement	Priority	Verify	Parent
FR-042	The system shall provide a Retention Health dashboard showing: account count by tier, ARR by tier, tier movement in the period, open cases by status, SLA compliance rate and save rate.	Must	D	BR-08
FR-043	The dashboard shall be filterable by CS team, CSM, plan tier, primary marketplace and date range.	Must	D	BR-08
FR-044	The dashboard shall present a Critical-tier worklist ordered by ARR descending, showing account, CSM, score, days in tier and case status.	Must	D	BR-08
FR-045	The dashboard shall show, for the selected period, the count of accounts whose opening tier was Critical split by final outcome — saved, churned, still open.	Must	D	BR-14
FR-046	The dashboard shall refresh at least once per day following completion of the scoring cycle, and shall display the last refresh timestamp.	Must	T	BR-08
FR-047	The system shall allow a user to export the current dashboard view to spreadsheet format, subject to the user's data access rights.	Should	D	BR-08

3.8 Administration and configuration — traces to BR-10

ID	Requirement	Priority	Verify	Parent
FR-048	The system shall allow a Business Administrator to maintain signal weights, and shall reject any set of weights that does not total 100%.	Must	T	BR-10
FR-049	The system shall allow a Business Administrator to maintain tier boundaries, hysteresis buffer and SLA durations per tier.	Must	T	BR-10
FR-050	The system shall allow a Business Administrator to maintain the risk-driver-to-playbook mapping and the case closure reason list.	Must	T	BR-10
FR-051	The system shall version every configuration change with the effective date, prior value, new value and the user who made it, and shall apply changes from the next scoring cycle onwards without altering historical scores.	Must	T	BR-10, BR-12
FR-052	The system shall provide a preview that applies a proposed configuration to the most recent cycle's data and reports the resulting tier distribution and alert volume, without saving.	Should	D	BR-10, R-3

3.9 Override, access and audit — traces to BR-12, BR-13

ID	Requirement	Priority	Verify	Parent
FR-053	The system shall allow a CSM to override an account's risk tier, requiring a reason from a controlled list, free-text justification and an expiry date not more than 90 days ahead.	Should	T	BR-13
FR-054	The system shall display the overridden tier alongside the calculated tier, never replacing it, and shall exclude overridden accounts from alert generation for the override period.	Should	D	BR-13
FR-055	The system shall expire an override automatically on its expiry date and resume normal alerting from the next cycle.	Should	T	BR-13
FR-056	The system shall restrict visibility of account risk data to the account's owning CSM, that CSM's management chain, CS Operations, and roles explicitly granted portfolio access.	Must	T	BR-12
FR-057	The system shall write an immutable audit entry for every configuration change, tier override, case closure and manual rescore, capturing user, timestamp, before value and after value.	Must	I	BR-12
FR-058	The system shall make the audit log available to authorised users as a filterable report with export.	Must	D	BR-12

4. Business rules

ID	Rule
BRule-01	An account is in scope for scoring where subscription status is Active and segment is High-Value.
BRule-02	Accounts with a cancellation already submitted are excluded from scoring and from alerting.
BRule-03	The composite score is always expressed 0-100; a higher score means higher churn risk.
BRule-04	Signal weights must total exactly 100%.
BRule-05	SLA duration is measured in business days on the CS working calendar, excluding public holidays configured per region.
BRule-06	An account newly onboarded within 60 days is scored but excluded from Critical alerting, as usage baselines are not yet meaningful.
BRule-07	A Closed-False Positive case must record which signal produced the false reading; these are reviewed in the quarterly model review.
BRule-08	A tier override never changes the stored calculated score or the score history.

5. Requirements coverage summary

Business requirement	Functional requirements
BR-01	FR-011, FR-014, FR-015
BR-02	FR-001 – FR-005, FR-010
BR-03	FR-016, FR-017, FR-018
BR-04	FR-021 – FR-026, FR-028, FR-029
BR-05	FR-013, FR-027
BR-06	FR-012, FR-022, FR-031, FR-032
BR-07	FR-035 – FR-041
BR-08	FR-042 – FR-047
BR-09	FR-019, FR-020, FR-033
BR-10	FR-048 – FR-052
BR-11	FR-006 – FR-009, FR-034
BR-12	FR-051, FR-056, FR-057, FR-058
BR-13	FR-053, FR-054, FR-055
BR-14	FR-041, FR-045
BR-15	Deferred — R2

No business requirement in R1 scope is unmapped. Full forward and backward traceability is held in [05_RTM_ChurnGuard.xlsx](#).